

Optimization – 2

Pointwise optimisation, extreme points and the bang-bang principle

10 сентября 2026 года

Theoretical minimum.

Def. A point of a convex set K is *extreme* if it is not the midpoint of a segment with both ends in K .

Theorem 1 (extremum at an extreme point). A linear functional on a compact convex set attains both its maximum and its minimum at extreme points; a convex continuous functional attains its maximum at an extreme point. By the Krein–Milman theorem a compact convex set is the closed convex hull of its extreme points, so extreme points always suffice.

Theorem 2 (extreme points of the sets one meets). The extreme points of the simplex are its vertices; of the ball $\{|f| \leq 1\}$ in L^∞ , the functions with $|f| = 1$ almost everywhere; of $\{0 \leq f \leq 1, \int f = c\}$, the indicators of sets of measure c ; of the doubly stochastic matrices, the permutation matrices (Birkhoff). Hence the answer to a linear extremal problem is typically a ± 1 -valued function or a permutation — this is the *bang-bang* form.

Fact 1 (the pointwise principle). If $J(f) = \int \Phi(x, f(x)) dx$ and the class of admissible f imposes no relation between values at different points, then $\sup J = \int \max_t \Phi(x, t) dx$: maximise inside the integral. One integral constraint costs one multiplier, and for Φ linear in t the maximiser is again bang-bang, with the switching point determined by the constraint.

-
1. Prove Theorem 1, describe the extreme points listed in Theorem 2, and prove Birkhoff's theorem. Deduce that the maximum of a linear functional over doubly stochastic matrices is attained at a permutation matrix.

2. (IMC) Ivan the Confessor prefers continuous functions $f: [0, 1] \rightarrow \mathbb{R}$ satisfying $f(x) + f(y) \geq |x - y|$ for all $x, y \in [0, 1]$. Find the minimum of $\int_0^1 f$ over all preferred functions.
3. (Proposed problems) Let $f: [0, 2\pi] \rightarrow \mathbb{R}$ satisfy $\int_0^{2\pi} f(x) \cos(kx) dx = 1$ for $k = 1, 2, \dots, n$, where $n \geq 2$. Find the minimum of $\int_0^{2\pi} f^2(x) dx$ over all such functions.
4. (S.T. Yau CSMC 2019) Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be strictly convex and let $u: [0, 1] \rightarrow \mathbb{R}$ be continuous with $\int_0^1 u = 0$. Show that $\int_0^1 F(u(x)) dx \leq \frac{F(\|u\|_\infty) + F(-\|u\|_\infty)}{2}$ and determine when equality occurs.
5. Prove the bang-bang solution of the problem: maximise $\int_0^1 w(x)f(x) dx$ over $|f| \leq 1$ with $\int_0^1 f = 0$, for a continuous strictly monotone w . Then recover from it the answer to Putnam 2011 from Optimization – 1 without any computation.